```
      const squawk=a.squawk?String(a.squawk).padStart(4,"0"):"????";
      const st=flightStatus(vsFpm);
      const infoEntry=state.info.get(a.icaoAddress);
      const info=infoEntry&&infoEntry.status==="done"?infoEntry.data:null;
      const infoPending=!infoEntry||infoEntry.status==="pending";
      const typeBits=info?[[info.manufacturer,info.type].filter(Boolean).join(" "),info.registration?`Reg ${info.registrat
 | ion}`:''].filter(Boolean):[];
      const subParts=[`ICAO24 ${a.icaoAddress}`,`Squawk ${squawk}`,...(typeBits.length?typeBits:(a.sourceIp?[`Feed ${a.sou
 | rceIp}`]:[]))];
      if(a.isPredicted)subParts.push(`Pred. age ${Math.round(a.predictionAgeSec)}s`);
      const subLine=subParts.join("  ·  ");
      let thumbHtml;
      if(info&&info.photoUrl){
        thumbHtml=`<a href="${info.photoLink||'#'}" target="_blank" rel="noopener"><img src="${info.photoUrl}" alt="${a.ca
 | llsign} photo" loading="lazy"/></a><div class="fthumb-credit">${info.photoCredit||"planespotters.net"}</div>`;
      }else if(infoPending){
        thumbHtml=`<div class="fthumb-ph">Looking up photo?</div>`;
      }else{
        thumbHtml=`<div class="fthumb-ph">No photo available</div>`;
      }
      const l=state.layers.get(a.icaoAddress);const live=l?l.tr.getLatLngs():[];const pred=l?l.pr.getLatLngs():[];const al
 | l=[...live,...pred].map(p=>[p.lat,p.lng]);let dist=0;for(let i=1;i<all.length;i++)dist+=haversine(all[i-1],all[i]);const
 | first=l&&l.origin?l.origin:(all.length?all[0]:[a.latDD,a.lonDD]),last=all.length?all[all.length-1]:[a.latDD,a.lonDD];con
 | st observedStart=nearestPlace(first[0],first[1]),observedCurrent=nearestPlace(last[0],last[1]);
      ensureRouteInfo(a.callsign,a.icaoAddress);
      const routeEntry=state.routeInfo.get(a.callsign);
      const routeInfo=routeEntry&&routeEntry.status==="done"?routeEntry.data:null;
      const routePending=!routeEntry||routeEntry.status==="pending";
      const realOrigin=routeInfo&&routeInfo.origin;
      const realDest=routeInfo&&routeInfo.destination;
      const fromName=realOrigin?(realOrigin.municipality||realOrigin.name)+(realOrigin.iataCode?` (${realOrigin.iataCode})
 | `:""):(routePending?"Looking up...":"Route not available");
      const fromSub=realOrigin?realOrigin.name:(routePending?"Checking scheduled origin":"No verified origin");
      const toName=realDest?(realDest.municipality||realDest.name)+(realDest.iataCode?` (${realDest.iataCode})`:""):(route
 | Pending?"Looking up...":"Route not available");
      const toSub=realDest?realDest.name:(routePending?"Checking scheduled destination":"No verified destination");
      const toLabel="To";
      document.getElementById("flight").innerHTML=`
        <div class="fcard">
          <div class="ftoprow">
            <div class="fmeta">
              <div class="fhead">
                <div class="fcall">${a.callsign}</div>
                <div class="fbadge ${st.cls}">${st.arrow} ${st.label}</div>
                ${a.isPredicted?'<div class="fbadge fbadge-pred">Predicted</div>':''}
              </div>
              <div class="fsub">${subLine}</div>
            </div>
            <div class="fthumb">${thumbHtml}</div>
          </div>
          <div class="stat-grid">
            <div class="stat"><div class="sk">Altitude</div><div class="sv">${altFt.toLocaleString()} ft</div><div class="
 | su">${altM.toLocaleString()} m</div></div>
            <div class="stat"><div class="sk">Ground speed</div><div class="sv">${spdKt} kt</div><div class="su">${spdMs}
 | m/s</div></div>
            <div class="stat"><div class="sk">Vertical speed</div><div class="sv">${vsFpm>0?'+':''}${vsFpm} ft/min</div><d
 | iv class="su">${vsMs.toFixed(1)} m/s</div></div>
            <div class="stat"><div class="sk">Heading</div><div class="sv">${hdg}°</div></div>
            <div class="stat"><div class="sk">Position</div><div class="sv">${a.latDD.toFixed(3)},
 | ${a.lonDD.toFixed(3)}</div></div>
            <div class="stat"><div class="sk">Distance flown</div><div class="sv">${(dist/1000).toFixed(0)} km</div></div>
          </div>
          <div class="routebar">
            <div class="pt from"><div class="pn">${fromName}</div><div class="pd">${fromSub}</div></div>
            <div class="arrow">&#9992;</div>
            <div class="pt to"><div class="pn">${toName}</div><div class="pd">${toSub}</div></div>
          </div>
        </div>`;
      const typeRows=info?`<dt>Aircraft type</dt><dd>${[info.manufacturer,info.type].filter(Boolean).join(" ")||"Unknown"}
 | ${info.icaoTypeCode?` (${info.icaoTypeCode})`:''}</dd><dt>Registration</dt><dd>${info.registration||"Unknown"}</dd><dt>C
 | ategory</dt><dd>${info.category}</dd>`:'';
      document.getElementById("aircraft").innerHTML=`<dl class="kv"><dt>ICAO24</dt><dd>${a.icaoAddress}</dd><dt>Callsign</
 | dt><dd>${a.callsign}</dd>${typeRows}<dt>Latitude</dt><dd>${a.latDD.toFixed(6)}</dd><dt>Longitude</dt><dd>${a.lonDD.toFix
 | ed(6)}</dd><dt>Altitude</dt><dd>${altFt.toLocaleString()} ft (${altM.toLocaleString()} m)</dd><dt>Ground speed</dt><dd>$
 | {spdKt} kt (${spdMs} m/s)</dd><dt>Vertical speed</dt><dd>${vsFpm} ft/min</dd><dt>Heading</dt><dd>${hdg}°</dd><dt>Squawk<
 | /dt><dd>${squawk}</dd><dt>Source feed</dt><dd>${a.sourceIp||"?"}</dd></dl>`;
      const airlineRow=routeInfo&&routeInfo.airlineName?`<dt>Airline</dt><dd>${routeInfo.airlineName}</dd>`:'';
      const routeStatus=realOrigin||realDest?"Confirmed by route database":(routePending?"Lookup in progress":"Not availab
 | le - no verified schedule found");
      document.getElementById("route").innerHTML=`<dl class="kv">${airlineRow}<dt>Route status</dt><dd>${routeStatus}</dd>
 | <dt>From</dt><dd>${fromName}</dd><dt>${toLabel}</dt><dd>${toName}</dd><dt>Observed start</dt><dd>${observedStart?`${obse
 | rvedStart.name} area (${observedStart.distanceKm.toFixed(0)} km)`:"Unknown"}</dd><dt>Current area</dt><dd>${observedCurr
 | ent?`${observedCurrent.name} area (${observedCurrent.distanceKm.toFixed(0)} km)`:"Unknown"}</dd><dt>Start Coord</dt><dd>
 | ${first[0].toFixed(6)}, ${first[1].toFixed(6)}</dd><dt>Current Coord</dt><dd>${last[0].toFixed(6)},
 | ${last[1].toFixed(6)}</dd><dt>Track Distance</dt><dd>${(dist/1000).toFixed(2)} km</dd></dl>`;
    }
    function pushAlt(a){const h=state.hist.get(a.icaoAddress)||[];h.push({t:new Date(),v:Math.round(a.altitudeMM/1000)});i
 | f(h.length>60)h.shift();state.hist.set(a.icaoAddress,h);if(state.sel!==a.icaoAddress)return;chart.data.labels=h.map(p=>p
 | .t.toLocaleTimeString("en-IN",{hour12:false,minute:"2-digit",second:"2-digit"}));chart.data.datasets[0].data=h.map(p=>p.
 | v);chart.update("none")}
    function animateAircraft(l,target,heading){if(l.anim)cancelAnimationFrame(l.anim);const start=l.m.getLatLng(),jump=hav
 | ersine([start.lat,start.lng],target);if(jump>100000){l.m.setLatLng(target);l.m.setRotationAngle(heading);l.heading=headi
 | ng;l.anim=null;return}const startHeading=Number.isFinite(l.heading)?l.heading:heading,delta=((heading-startHeading+540)%
 | 360)-180,started=performance.now(),duration=1250;const step=(now)=>{const t=Math.min(1,(now-started)/duration);l.m.setLa
 | tLng([start.lat+(target[0]-start.lat)*t,start.lng+(target[1]-start.lng)*t]);l.m.setRotationAngle(startHeading+delta*t);i
 | f(t<1){l.anim=requestAnimationFrame(step)}else{l.anim=null;l.heading=heading}};l.anim=requestAnimationFrame(step)}
    function ensure(a){let l=state.layers.get(a.icaoAddress),hd=a.headingDE2/100;if(!l){const cachedInfo=infoFor(a.icaoAdd
 | ress);const m=L.marker([a.latDD,a.lonDD],{icon:iconFor(cachedInfo&&cachedInfo.category),rotationAngle:hd}).addTo(map);m.
 | bindTooltip(a.callsign,{permanent:true,direction:"right",offset:[10,0],className:"lbl"}).openTooltip();m.bindPopup(pop(a
 | ));m.on("click",()=>select(a.icaoAddress));ensureInfo(a.icaoAddress);const tr=L.polyline([],{color:"#8ecae6",weight:2,op
 | acity:.5}).addTo(map),pr=L.polyline([],{color:"#ff5d5d",weight:2,dashArray:"7 7",opacity:.95}).addTo(map),ve=L.polyline(
 | [],{color:"#bde0fe",weight:1,dashArray:"6 6"}).addTo(map);l={m,tr,pr,ve,data:a,vis:true,origin:[a.latDD,a.lonDD],heading
 | :hd,anim:null};state.layers.set(a.icaoAddress,l)}l.data=a;animateAircraft(l,[a.latDD,a.lonDD],hd);l.m.setPopupContent(po
 | p(a));l.m.setTooltipContent(a.callsign+(a.isPredicted?" PRED":""));if(a.isPredicted){l.pr.addLatLng([a.latDD,a.lonDD]);c
 | onst pPts=l.pr.getLatLngs();if(pPts.length>120)l.pr.setLatLngs(pPts.slice(-120));}else{l.tr.addLatLng([a.latDD,a.lonDD])
 | ;const pts=l.tr.getLatLngs();if(pts.length>120)l.tr.setLatLngs(pts.slice(-120));if(l.pr.getLatLngs().length>0)l.pr.setLa
 | tLngs([]);const dist=(a.horVelocityCMS/100)*20,rad=hd*Math.PI/180,dLat=(dist/111320)*Math.cos(rad),scale=Math.max(.2,Mat
 | h.cos(a.latDD*Math.PI/180)),dLon=(dist/(111320*scale))*Math.sin(rad);const vColor=a.isPredicted?"#ff5d5d":"#bde0fe";cons
 | t vDash=a.isPredicted?"4 6":"6 6";l.ve.setStyle({color:vColor,dashArray:vDash});l.ve.setLatLngs([[a.latDD,a.lonDD],[a.la
 | tDD+dLat,a.lonDD+dLon]]);pushAlt(a)}
    function showLayer(l,v){if(v&&!l.vis){l.m.addTo(map);l.tr.addTo(map);l.pr.addTo(map);l.ve.addTo(map);l.vis=true}else i
 | f(!v&&l.vis){map.removeLayer(l.m);map.removeLayer(l.tr);map.removeLayer(l.pr);map.removeLayer(l.ve);l.vis=false}}
    function cleanup(active){for(const [i,l] of state.layers){if(active.has(i))continue;if(l.anim)cancelAnimationFrame(l.a
 | nim);map.removeLayer(l.m);map.removeLayer(l.tr);map.removeLayer(l.pr);map.removeLayer(l.ve);state.layers.delete(i);state
 | .hist.delete(i);if(state.sel===i){state.sel=null;clearRoute();document.getElementById("flight").innerHTML='<div class="e
 | mpty">Selected aircraft left the feed.</div>';document.getElementById("aircraft").innerHTML='<div class="empty">No aircr
```